

WASP-43b

R-0607

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_180325 · created 2026-07-25T06:26:50+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458202.7554 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.107 +/- 0.052
Transit depth (Rp/Rs)^2	1.14 +/- 1.1 [%]
Orbital Inclination (inc)	86.67 +/- 2.9
Airmass coefficient 1 (a1)	0.986 +/- 0.043
Airmass coefficient 2 (a2)	0.009 +/- 0.067
Scatter in the residuals of the lightcurve fit is	4.37 %
Best Comparison Star	#2 - [350, 77]
Optimal Aperture	2.41
Optimal Annulus	9.24
Transit Duration (day)	0.0556 +/- 0.0054

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.107 ± 0.052 vs 0.1594 (deviation 1.01σ)
Duration fit vs published	0.0556 d vs 0.0512 d (deviation 0.81σ)
Mid-time O–C	-3.7 min
Depth SNR	2.1
Residual scatter	4.37 %

Light curve

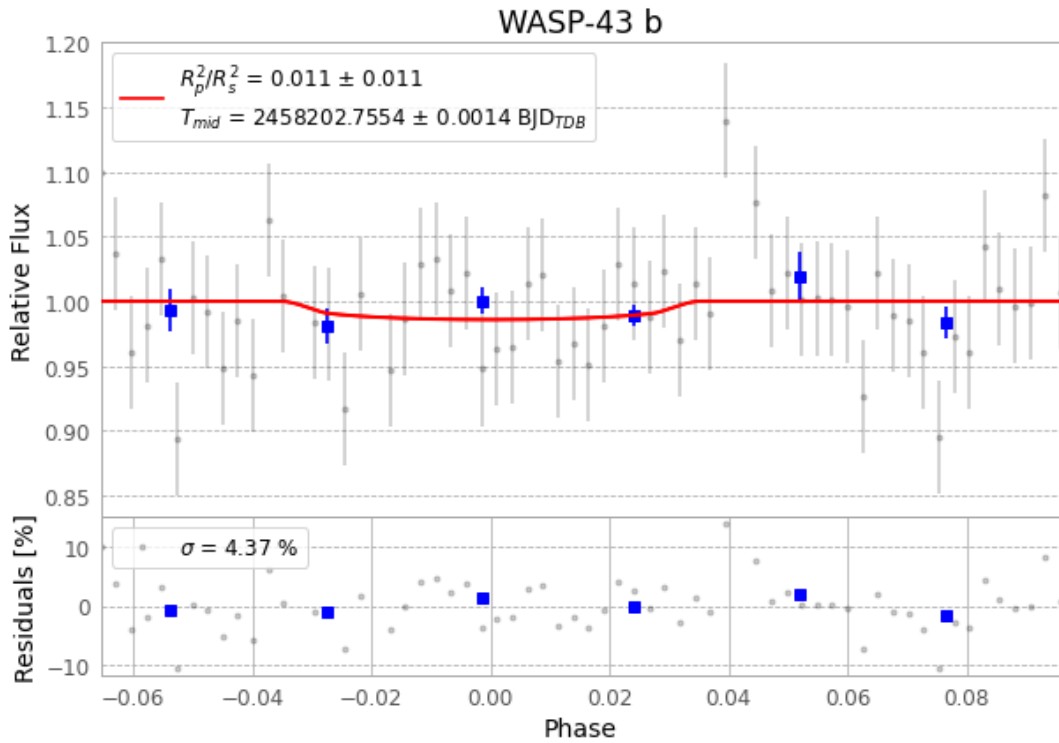


Figure 1 — FinalLightCurve_WASP-43 b_2018-03-24.png (SHA-256 e8bae21ef446dc16...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [427, 271], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 10edea8495aa0c12...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit b365494

Data provenance

64 FITS frames (42 MB), WASP-43180325044212.FITS → WASP-43180325075112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-180325.log (d6c9621c6c61...), FinalLightCurve_WASP-43 b_2018-03-24.png (e8bae21ef446...), FinalLightCurve_WASP-43 b_2018-03-24.csv (14484030cdc1...)

Compute resources

Wall time 120.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-25 06:26Z.